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Module 6

CompTIA Security+ Exam Objectives

Domain	**Lab**
Threats, Vulnerabilities, and Mitigation	6.1, 6.2
Security Architecture	6.2, 6.4
Security Operations	6.1, 6.2, 6.3, 6.4, 6.5

Website links change frequently. If a link in the following activities no longer works, use a search engine such as Google or Bing to search for the information.

Lab 6.4 Setting Up Physical Security of Mobile Devices

Estimated Time: 30 minutes

Objectives

In the realm of cybersecurity, physical security is generally overlooked for mobile devices. Physical security can be as simple as installing cables that attach a laptop to a desk or using password-locking screens. It can also involve armed security guards at doorways or software installed on devices.

After completing this lab, you will be able to do the following:

- Identify basic physical protections available to users of mobile devices.
- Identify that employee training should be a part of physical security.

Certification Objectives

Objectives for the CompTIA Security+ exam:

- 3.1 Compare and contrast security implications of different architecture models.

- 4.1 Given a scenario, apply common security techniques to computing resources.

Materials Required

This lab requires the following:

A computer with Internet access running the Windows operating system

Activity

In this lab, you learn about the importance of physical security.

1. Open a web browser and navigate to **https://www.sans.edu/cyber-research/37120/**.

2. Download the article titled "Physical Security and Why It Is Important." Read the Introduction section (section 1), which details the physical security of USB hard drives, laptops, tablets, and smartphones. For example, since mobile devices can be easily lost or stolen, cable locks and/or RFID systems can be put in place to help stop or diminish the potential loss.

3. Read section 2, "Planning For a Physical Security Program," and pay special attention to the "Defense in Depth" section.

4. Write a one-page paper that includes your definition of defense in depth. Also include a diagram based on your understanding of the concept.

5. Read the rest of the article and save it for future reference.

Defense in Depth in Physical Security

As Stephanie Nappo once stated, "it takes 20 years to build the reputation of a business and 20 minutes of one cyber-incident to destroy it." Many companies, even today, are figuring out the hard way how easy it is for hackers and threat actors to find their way onto compromised systems. This can be achieved through cell phones, laptops, USB devices, and tablets, posing the risk of data theft or misuse. This is directly affected by the sheer amount of cell phones and laptops that are stolen each day. According to the SANS *Physical Security and Why It Is Important (2021),* physical security must have a multi-layered approach using structure and over-lapping security features to protect individuals, facilities, equipment, and information of a company.

One of the most important principles supporting this approach is **defense in depth**.

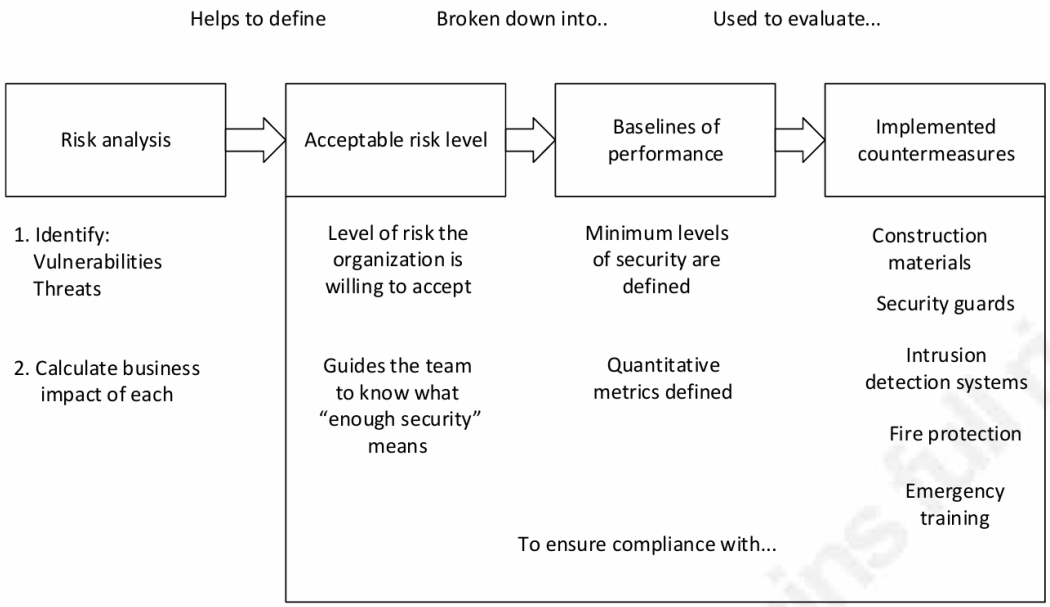
Defense in depth refers to the multi-layered security approach of utilizing multiple overlapping control systems to protect assets, in case one system fails, others continue to provide support. This ensures no single point failure in the security system. Defense in depth uses the knowledge that attackers may get past one or two of the layers as no system is impenetrable. However, the attackers must bypass seven or eight layers before accessing critical information or resources. This strategy is used to deter, deny, detect, and delay any attacker long enough for the company to respond and prevent the attack as its ongoing.

The SANS article states that defense in depth comes from not just one system, but utilizes a combined approach of using administrative, technical, and physical controls for security reinforcement to give a company true defense in depth.

For example, a pen tester attempting to access a company's secured computer system, they would encounter perimeter fences, security guards and a guard post. If they managed to access the perimeter, they would still need badges, smart cards, or other means to crack the door locks and fool the security cameras and intrusion detection systems. All of these provide additional layers of security. Lastly, smart card authentication and passwords use system-level protection for data. Each layer added to the defense in depth increases the difficulty of being hacked, and therefore lowers the risks associated with being hacked. That is not to say the system is by any means impenetrable, it just means it's more secure than systems with only one or two layers.

As increased communication and business are conducted on mobile and portable devices, defense in depth is more important than ever on said devices. We must understand that these devices are stolen so often that only multiple safeguards can protect the (hopefully) encrypted information stored on it. Companies use cable locks, RFID tracking chips, secure storage rooms, access controls, and monitoring systems to watch for anomalies of network access or data access. Defense in depth shows that even if one system fails, such as a stolen work phone out of a company vehicle, the other layers like end-to-end encryption, access logging, and other defenses can reduce the impact of the stolen cellular device. To improve resilience against internal and external threats, defense in depth states that no single security measure is safe and must be multi-layered.

In conclusion, defense in depth strengthens physical security by ensuring that assets are protected through multiple, coordinated layers. This approach reduces risk, increases the likelihood of detecting intrusions, and helps organizations protect people, information, and infrastructure even when individual controls are compromised.



Relationships of risk, baselines and countermeasures Harris, (2013)